

Bachelor in Computer Applications

Course Title: **C Programming**

Code No: Semester: **1**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. List the advantages and disadvantages of global variable. Assume the two data files "num1.txt" and "num2.txt" containing integers. Merge these two files to the third file named " file3.txt". Here, the integers of " file3.txt" must be written in sorted order.
2. How nested Structure is defined and initialized? Explain with an example. Write a program, using structure, to input records of 10 students. Members include name, roll number, marks obtained in math, C program and English. Display the records of students who have passed in C Program.
3. When do we use for loop and while loop? Compare the while and do while loop with suitable example. Write a program to find average, minimum and maximum age in a class of 20 students.
4. What do you mean by Flowchart? Draw a Flowchart to find the largest among three numbers entered by user.
5. Why array and pointer is similar? Write a program to show the similarity between an array and pointer.
6. Write a program to find the sum of diagonal elements of $n \times n$ matrix.
7. How keyword is different from Variables? Discuss Basic data types with its ranges.
8. What is operator associativity? Explain conditional and Logical operators in brief.
9. Write a program with a function that takes an array as an argument and returns its sum to the main function.
10. How do you include a file to another file? Illustrate with an example.
11. Write a program to display the following pattern using the string "NEPAL"NNENEPNEPANEPAL
12. Write short notes on:
a. Macro vs Function
b. Break vs Continue